

$$T=12$$

1. If the signal $x[n] = 8 \cos(\frac{\pi}{6}n + \frac{\pi}{2})$ is obtained through the 9.6KHz sampling, then the true frequency of the original analog signal is 800 Hz

2. $x[n]$ is a finite length sequence, $x[n] = \{2, 3, 1, 4, -3, -1, 1, 0, 6\}$, $X[k]$ is the 10 points DFT of this sequence, so $X[0] = \underline{14}$ $X[k] = \sum_{n=0}^9 x[n] e^{-j2\pi kn/10}$

3. A 9 points average filter is described by $y[n] = \frac{1}{9} \sum_{l=0}^{8} x[n-l]$, please find out its group delay $\tau(\omega) = \underline{4}$ $\sum_{l=0}^8 x[n-l] h[l] \quad h[l] = \frac{1}{9}$

4. If a finite length sequence $\{x[n]\} = \{1, 2, 2, 3, 3\}$, then $x[\langle 4-n \rangle_5] = \underline{\{9, 7, 5, 3, 1\}}$

5. $\{X_1[k]\}$ is obtained by the 6-point DFT transformation of the sequence

$$\{x[n]\} = \{1, 2, 2, 3, 3\} \quad \text{Apply the IDFT to } \{X_1[k]\}, \text{ we get}$$

$$\{y_1(n)\} = \underline{\{1, 2, 2, 3, 3\}} \quad \{X_2[k]\} \text{ is obtained by the 3-point DFT of}$$

$$\{x[n]\}. \text{ Apply the IDFT to } \{X_2[k]\}, \text{ we get } \{y_2(n)\} = \underline{\{6, 5, 2\}}$$

6. Given a 9-point real sequence with DFT values of $X[0] = 23, X[1] = 2.2426 - j,$

$$X[4] = -6.3749 + 4.1212j, X[6] = 6.5 + 2.5891j, X[7] = -4.1527 - 0.2645j \text{ at five}$$

points, determine the DFT values at the remaining 2 points:

$$X[2] = \underline{-4.1527 + 0.2645j}, X[8] = \underline{2.2426 + j}$$

$$X[2] = X^*[8] = X^*[7] \quad X[8] = X^*[1] = X^*[2]$$

7. Let the sequence $x[n]$ of length N be padded with $(M-1)N$ zeros, resulting in

an MN -point DFT $Y[k]$. Prove that the N-point DFT $X[k]$ of $x[n]$ can be

obtained from $Y[k]$ by the following formula.

$$X[k] = Y[Mk], \quad 0 \leq k \leq N-1$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

$$Y[k] = \sum_{n=0}^{MN-1} y[n] e^{-j2\pi kn/MN}$$

$$\text{Let } k = Mk, Y[Mk] = \sum_{n=0}^{MN-1} x[n] e^{-j2\pi Mk n/MN} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} = X[k]$$

8. A known sequence

$$x_{cs}(n) = \frac{x(n) + x^*(L-n)}{2}$$

$$= \{-1, 3.5+j, -2+j1.5, 2, -3-j2.5, 3.5-j\}$$

$$X_{ca}(n) = \frac{x(n) - x^*[c-n]}{2}$$

$$= \{-j7, -0.5, 5+j4.5, j^2, -j+j4.5, 0.5\}$$

$$x(n) = \{\underbrace{-1-j7}_0, \underbrace{3+j}_1, \underbrace{2+j7}_2, \underbrace{2+j2}_3, \underbrace{-8+j2}_4, \underbrace{4-j}_5\} \quad 0 \leq n \leq 5, \quad N=6$$

its FFT is $X[k]$. Write down its circularly conjugate symmetry $x_{cs}(n)$ and circularly conjugate antisymmetry $x_{ca}(n)$, and use $X[k]$ to indicate their FFT.

9. Let $X[k], 0 \leq k \leq 11$, be a 12-point DFT of a length-12 real sequence $x[n]$ with the first seven samples of $X[k]$ given by

$$X(k) = \{12, -18-j21, -10+j4, -6+j7, 9+j8, 19-j16, 39\} \quad 0 \leq k \leq 6$$

- 1) Determine the remaining samples of $X[k]$.

- 2) Evaluate the following functions of $x[n]$ without computing the IDFT of

$$X[k]: \text{(a) } x[6] \quad \text{(b) } \sum_{n=0}^{11} x[n] \quad \text{(c) } \sum_{n=0}^{11} e^{j\pi n/3} x[n]$$

$$\begin{aligned} 1) \quad X[7] &= X[-5] = X^*[5] = 19+j16 \\ X[8] &= X[-4] = X^*[4] = 9-j8 \\ X[9] &= X[-3] = X^*[3] = -6-j7 \\ X[10] &= X[-2] = X^*[2] = -10-j4 \\ X[11] &= X[-1] = X^*[1] = -18+j21 \end{aligned}$$

$$\begin{aligned} 2) \quad (a) \quad X[6] &= \frac{1}{12} \sum_{n=0}^{11} x[n] e^{-j\pi n} = \frac{1}{12} \sum_{n=0}^{11} x[n] (-1)^n \\ &= \frac{59}{12} = 4.917 \end{aligned}$$

$$(b) \quad k=0, \sum_{n=0}^{11} x[n] = X[0] = 12$$

$$(c) \quad \sum_{n=0}^{11} x[n] e^{j\frac{2\pi}{12} n \cdot (-4)}, \quad k=-2$$

$$X[-2] = X[10] = -10-j4$$